

Pattern Matching and Machine Learning for Audio Signal Processing

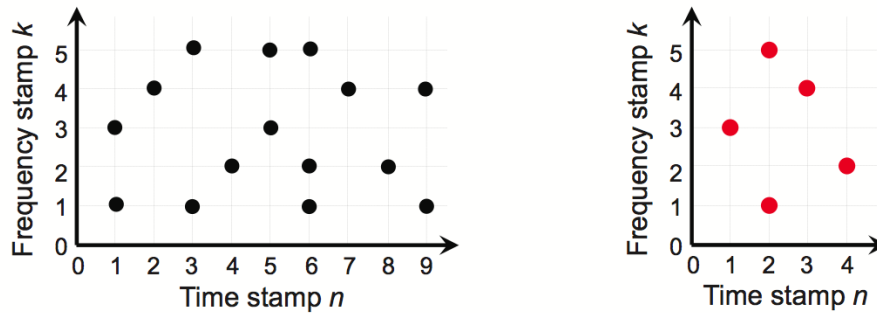
Exercise sheet 7

To be uploaded in eCampus till: 27-06-2026 22:00 (strict deadline)

Exercise 7.1

[3 + 4 + 2 = 9 points]

Consider the constellation maps $C(\mathcal{D})$ (left) and $C(\mathcal{Q})$ (right) as specified by the following figure.



- Write both $C(\mathcal{D})$ and $C(\mathcal{Q})$ as sets of (time stamp, frequency stamp) pairs.
- Compute the shifted versions $m + C(\mathcal{Q})$, for $m \in [-3 : 8]$.
- Determine the resulting matching function $\Delta_C : \mathbb{Z} \rightarrow \mathbb{N}_0$ defined by

$$\Delta_C(m) := |(m + C(\mathcal{Q})) \cap C(\mathcal{D})|.$$

Exercise 7.2

[5 + 5 = 10 points]

Let $C(\mathcal{D})$ and $C(\mathcal{Q})$ be defined as in Exercise 7.1.

- Compute the corresponding inverted lists and the indicator functions as used in the lecture.
- Compute the resulting matching function.